

Customer Shopping Behavior Analysis

End-to-end customer analytics project report

Scope	Prepared from the supplied project portfolio material
Dataset	3,900 retail purchase records and 18 source columns
Analytics stack	Python, Pandas, MySQL SQL, and Power BI
Report purpose	Summarize customer behavior findings, methods, limitations, and next actions

This report documents an analytics workflow that prepares retail purchase data in Python, answers business questions in SQL, and communicates results in Power BI. Figures and observations are limited to the supplied project material.

Executive Summary

The analysis shows that category mix, customer age, and repeat-purchase history are the clearest commercial patterns in the supplied dataset. Clothing is the largest category, Young Adults produce the highest revenue and order volume, and Loyal customers make up the largest customer segment. Subscription status explains customer count and total revenue more than transaction value: non-subscribers generate more revenue because they are the larger group, while average spend is nearly identical between the two groups.

\$233,081	3,900	\$59.76	3.75
Total purchase value	Customer records	Average purchase amount	Average review rating

Headline Findings

- **Category:** Clothing contributes about \$104K of revenue and 44.54% of orders, clearly ahead of the other categories.
- **Age:** Young Adults lead revenue at \$62,143 and orders at 1,028.
- **Retention:** 3,116 customers are Loyal, defined as having more than 10 previous purchases.
- **Subscription:** Subscribers are 27% of customers. Their average spend is \$59.49 versus \$59.87 for non-subscribers.
- **Fulfilment:** Express orders average \$60.48, moderately above Standard orders at \$58.46.

Recommended Focus

Prioritize Clothing merchandising and Young Adult engagement; preserve and learn from the Loyal segment; test subscription benefits that improve frequency or retention rather than assuming higher spend per order; and use product-level discount and rating patterns to guide promotions and assortment decisions.

Project Scope and Data

The source data includes 3,900 customer purchase records with demographic, product, transaction, fulfilment, promotion, subscription, and purchase-history attributes. The project is designed to answer decision-oriented questions about revenue, behavior, products, and customer segments.

Core Data Fields

Domain	Representative fields	Analytical use
Customer	Customer ID, Age, Gender, Location	Demographic and geographic comparisons
Product	Item Purchased, Category, Size, Color	Category and product performance
Transaction	Purchase Amount, Season, Payment Method	Revenue and buying context
Relationship	Subscription Status, Previous Purchases, Frequency	Repeat buyer and loyalty analysis
Experience	Review Rating, Shipping Type	Product sentiment and fulfilment analysis
Promotion	Discount Applied, Promo Code Used	Promotion uptake assessment

Data Preparation and Workflow

Python was used to inspect the CSV, fill missing review ratings by category median, standardize column names to lowercase snake case, derive age groups with quartiles, map buying-frequency labels to days, and remove the redundant promo-code field. The prepared data was exported to MySQL for SQL analysis, then summarized in a Power BI dashboard.

Stage	Output
Python and Pandas	Cleaned analytical dataset; 37 missing review ratings imputed by category median
MySQL SQL	Ten repeatable business questions using aggregations, subqueries, CTEs, CASE, and window functions
Power BI	KPI cards, slicers, subscription mix, and category and age-group comparisons

Important implementation note: the notebook exports a table named **customer**, while the SQL script queries **customers**. The table name must be aligned before the SQL queries can be reproduced.

Customer and Revenue Analysis

Revenue is concentrated in Clothing and is led by Young Adults. The findings describe the supplied transaction records and should be interpreted as descriptive, not causal.

Revenue by Age Group

Age group	Revenue	Orders
Young Adult	\$62,143	1,028
Middle Age	\$59,197	986
Adult	\$55,978	942
Senior	\$55,763	944

Category Performance

Category	Revenue	Order share
Clothing	About \$104K	44.54%
Accessories	About \$74K	31.79%
Footwear	About \$36K	15.36%
Outerwear	About \$19K	8.31%

Gender Revenue

Gender	Revenue
Male	\$157,890
Female	\$75,191

Subscription and Repeat Buyers

Status	Customers	Average spend	Total revenue	Repeat buyers
Subscriber	1,053	\$59.49	\$62,645	958
Non-subscriber	2,847	\$59.87	\$170,436	2,518

Interpretation: non-subscribers account for 73% of customers and therefore generate most revenue. The \$0.38 difference in average spend does not support a conclusion that subscription increases spend per transaction.

Product, Promotion, and Fulfilment Analysis

Product demand, customer ratings, and promotion uptake identify practical candidates for assortment, messaging, and controlled offer testing.

Top Products by Category

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Pants (171)	Shirt (169)	Blouse (158)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	-

Highest Rated Products

Product	Average rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Application Rates

Product	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping Comparison

Shipping type	Average purchase amount
Express	\$60.4752
Standard	\$58.4602

The higher Express average is a useful hypothesis for further testing, but the provided results do not establish whether shipping choice itself drives spend or merely correlates with it.

Segmentation, Actions, and Limitations

Previous-purchase history provides a simple, operational segment definition. Loyal customers dominate the base, while New and Returning customers represent smaller opportunities for activation and progression.

Customer Segments

Segment	Definition	Customers
New	Exactly 1 previous purchase	83
Returning	2 to 10 previous purchases	701
Loyal	More than 10 previous purchases	3,116

Recommended Actions

- **Protect Clothing momentum.** Prioritize availability and cross-sell paths around Pants and other leading Clothing items.
- **Tailor Young Adult engagement.** Use this group as the first audience for category and product experiments because it leads revenue and orders.
- **Build subscription experiments around behavior.** Test benefits designed to lift retention, frequency, or basket composition; track incremental effects against a comparison group.
- **Use discount discipline.** Review high-discount-use products such as Hat, Sneakers, and Coat for margin impact and incremental conversion.
- **Deepen retention analysis.** Apply RFM, cohort, and lifetime value methods once transaction dates and richer customer history are available.

Data and Reporting Limitations

- Thirty-seven review ratings were imputed with category medians; this preserves records but may reduce variation in rating analysis.
- No duplicate-removal step, notebook charts, statistical tests, joins, or saved notebook visualizations were present in the supplied files.
- The Power BI .pbix file was not supplied, so model relationships, refresh configuration, and dashboard calculations could not be confirmed.
- Database credentials appear directly in the notebook and should be moved to environment variables or a secrets manager.
- No transaction-date field was provided, preventing time-series, cohort, and seasonality validation beyond the supplied season label.

Conclusion

This project demonstrates a coherent customer analytics workflow from preparation through dashboard communication. Its strongest immediate messages are the central role of Clothing, Young Adult revenue leadership, the scale of the Loyal segment, and the need to evaluate subscription impact through carefully designed behavioral experiments rather than transaction-average comparisons alone.